Mechanical waves and distant effects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use travel-time reasoning while distinguishing waves from the motion of material.

SUCCESS CRITERIA - I can...

1. Identify the medium carrying the wave.
2. Use speed and distance to estimate travel time.
3. Explain why a simple travel time does not predict damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A bundle of parallel rays passes through a lens and becomes 3 times as wide farther beyond it, with rays spreading immediately after the lens. Identify the lens type and sketch the rays.

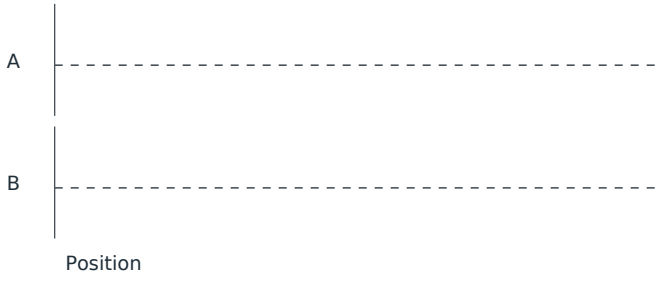
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $d = v\,\Delta t$; $\Delta t = d/v$.
Use the supplied wave speed for its medium and wave type. These classroom models are not real hazard forecasts.

Worked reasoning: In a simple model, a wave amplitude falls from 4 to 2 units while intensity is proportional to amplitude squared. Intensity becomes one quarter, not one half.

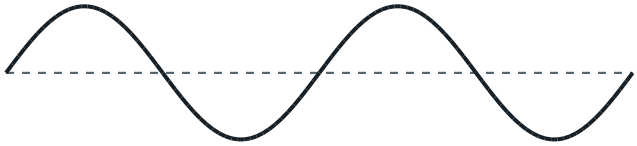
Your turn: Draw the two waves and two corresponding relative-intensity bars.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw a disturbance traveling through water and mark local water motion. Why does wave arrival not mean the original water crossed the whole ocean?



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

In a water-wave simulation, a disturbance travels at 20 m/s between sensors 200 m apart. Calculate arrival time. Mark the one-way path and explain why this is not the speed of an individual water particle.



Marzano check - circle: ① ② ③

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Continue down each column.

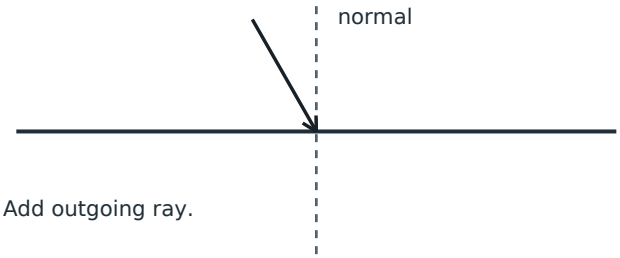
05 Explain your reasoning

Why does knowing wave arrival time alone not establish local damage? Name other evidence.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

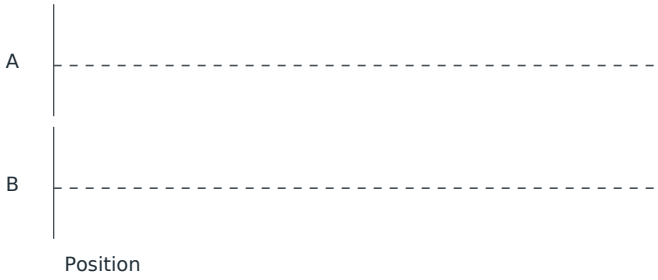
A wave enters a medium where its speed is 4 times as large, at a small angle below critical conditions. Does the transmitted ray bend toward or away from the normal? Label the media, normal and ray directions.



Marzano check - circle: 1 2 3

07 Exit check

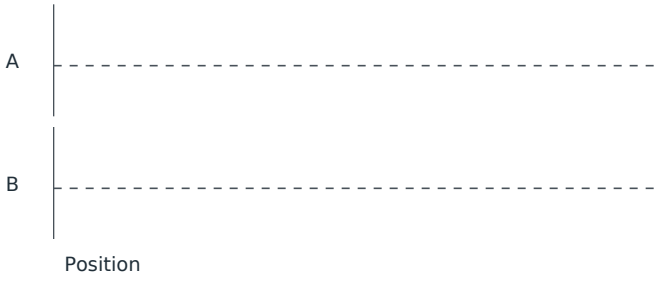
At fixed speed, distance doubles. How does travel time change?



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two sites have amplitudes 3 and 1 units. Compare relative intensities and name one real-world factor besides distance that could affect the readings.



Marzano check - circle: 1 2 3